Dynamic Arrest Follows Assembly on a Temperature-Scaled Common Clock in an Elastin-like Polypeptide

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Abstract

Many proteins assemble into dense phases that then lose their mobility, as when a condensate hardens or a protein gel sets. Assembly and arrest are usually followed with

different tools, so whether arrest tracks structure, and what sets its pace, have been difficult to establish. Small-angle X-ray photon correlation spectroscopy (SA-XPCS) records both structure and dynamics on the same detector frames. We apply it to the elastin-like polypeptide $(VPAVG)_{30}$ in isothermal experiments at four temperatures between 28.00 and 28.75 °C. The protein assembles at all four temperatures. Its correlation functions decay in two steps, and the amplitude of the fast step falls as the sample arrests. At 28.50 °C, this amplitude drops from 0.80 to 0.07 at 175 nm within an hour, and less far at every smaller length scale down to 83 nm. One rescaling of the waiting time, with a scaling temperature $T_s = 0.24 \pm 0.02$ K, collapses the arrest at all four temperatures onto a single master curve. Arrest thus speeds up by about a factor of three for every quarter of a degree of heating. The low-Q SAXS intensity follows the same clock. Arrest nonetheless trails assembly: every sample that arrests is half arrested at nearly the same low-Q intensity, after most of its low-Q scattering has developed. The steepness of the clock corresponds to an apparent activation energy more than a thousand times the thermal energy. It points to a rate-limiting step in which many contacts close together.

Significance Statement

Many proteins assemble into dense phases that later stop moving, as when a condensate hardens or a gel sets. Structure and motion are usually measured separately, so whether the loss of motion tracks the growth of structure has been hard to tell. We record both at once in a designed protein polymer with fast X-ray scattering. Both run on one clock, and the motion stops after most of the structure has formed. The clock runs about three times faster for every quarter of a degree of heating, so a fraction of a degree can determine when this protein material sets. This steepness points to a rate-limiting step in which many molecular contacts form together.

Introduction

Many proteins assemble into dense phases that progressively lose their mobility, a process known as dynamic arrest. Biomolecular condensates, for example, become more viscous as they age,[1] and some convert from liquid droplets into solid aggregates, a transition linked to neurodegenerative diseases such as amyotrophic lateral sclerosis.[2,3] Heat-set protein gels such as cooked egg white, and protein scaffolds that set at body temperature, follow a similar sequence: the proteins assemble first, and the network they form arrests later.[4–6] In each case, the properties that matter, from the viscosity of a condensate to the stiffness of a gel, depend on how far and how fast the assembly has arrested. Assembly and arrest are, however, usually followed with different tools. Scattering and microscopy record structure, whereas mobility is measured by microrheology, bulk rheology and, for condensates, fluorescence recovery after photobleaching, each at its own length scale.[7] Whether arrest simply tracks the growth of structure, and what sets its pace, have therefore been difficult to establish. The answers matter both for controlling when a protein material sets and for understanding how a condensate hardens.

Protein polymers whose assembly is encoded in their sequence are well suited to these questions, and elastin-like polypeptides (ELPs) are the best developed of them. ELPs are intrinsically disordered proteins whose sequence, molecular weight, and lower critical solution temperature (LCST) behavior can be programmed recombinantly,[8–12] and they have been proposed as minimal models of the disordered proteins that form condensates.[13,14] Above the LCST, ELPs undergo hydrophobic desolvation and can form coacervates or hierarchical assemblies depending on sequence, concentration, and thermal history.[11,12,15] In particular, alanine-containing ELPs can form stiff, thermoresponsive gels through arrested phase separation.[16,17] Although these studies establish the arrested state from time-averaged scattering and rheology, how an assembly evolves toward arrest, and what paces it, remains elusive.

Following an assembly toward arrest requires measuring its structure and its dynamics at

the same time, because the two can evolve on different time scales. Small-angle X-ray photon correlation spectroscopy (SA-XPCS) measures both simultaneously, on the same detector frames: averaging the frames in time gives the SAXS profile, and correlating them in time measures how fast that structure rearranges, at the same low-Q length scales. Measuring them together places assembly and arrest in the same sample volume, at the same moment and on the same length scales, so the order and relative pace of the two can be read directly. In protein systems, XPCS has resolved two relaxation modes with ballistic droplet motion during liquid–liquid phase separation,[18] temporally heterogeneous dynamics during gelation,[5] and stretched cage relaxation in crowded solutions.[19] A protein-rich morphology can likewise keep reorganizing through junction exchange, whose kinetics set the network relaxation time in associative networks,[20] and through the residual coarsening that follows arrested phase separation.[21,22] Both can proceed while the SAXS profile changes only modestly.

Here we use SA-XPCS to follow assembly and arrest together in concentrated aqueous $(\mathrm{VPAVG})_{30}$, an ELP built from 30 repeats of the pentapeptide valine-proline-alanine-valine-glycine. In isothermal experiments at 28.00, 28.25, 28.50 and 28.75 °C, the protein assembles at all four temperatures. The arrest that follows is strongly temperature dependent and sets in earlier at higher temperature. When the waiting time is rescaled with a single temperature scale, $T_s = 0.24 \pm 0.02$ K, the arrest at all four temperatures falls onto one master curve: arrest speeds up by about a factor of three for every quarter of a degree of heating. The growth of structure seen by SAXS follows the same temperature-scaled clock. Arrest nonetheless trails assembly. By the time a sample is half arrested, most of its low-Q scattering has already developed, and arrest continues while the SAXS intensity grows by less than a factor of two. This answers, for one protein polymer, the two questions raised at the outset: arrest trails assembly on a common clock, and temperature sets the pace of both. For protein materials, the time to arrest thus becomes a quantity that can be measured directly and tuned by sub-degree changes in temperature, and the steepness of its temperature scaling is a kinetic measure of how many contacts must form together in the step that sets it. For condensates,

whose disordered proteins phase separate much as ELPs do, a cooperative step that paces assembly and arrest alike may determine when they turn from liquid to solid.

Results

Fig. 1 summarizes the SA-XPCS experiment. All four loadings came from a single batch of $(\mathrm{VPAVG})_{30}$ at $101\,\mathrm{mg\,mL^{-1}}$ in $20\,\mathrm{mmol\,L^{-1}}$ HEPES, $250\,\mathrm{mmol\,L^{-1}}$ NaCl and $2\,\mathrm{mmol\,L^{-1}}$ DTT at pH 8. Each 2 s SA-XPCS acquisition at Advanced Photon Source (APS) beamline 8-ID-I (Materials and Methods) recorded 100,000 detector frames. Averaging the frames in time and azimuth yields $I(Q)$, and multi-tau correlation of the same frames yields the intensity autocorrelation function

$$g_2(Q,\Delta t) = \frac{\langle I(Q,t)I(Q,t+\Delta t)\rangle_t}{\langle I(Q,t)\rangle_t^2}, \tag{1}$$

where $I(Q,t)$ is the coherent scattering intensity and Δt is the delay time. Through the Siegert relation, g_2 reports the relaxation of density fluctuations at length scale $2\pi/Q$.

To resolve mesoscale arrest isothermally, each loading was equilibrated at 6 °C for at least 30 min and measured at that temperature. It was warmed from there to 20 °C at $1\,^\circ\mathrm{C\,min^{-1}}$ and heated at $0.1\,^\circ\mathrm{C\,min^{-1}}$ to a final temperature between 28.00 and 28.75 °C, where it was held. The first ramp ended at 20 °C because no appreciable assembly or dynamic arrest was observed at room temperature. The second ramp was kept slow to prevent the temperature from overshooting the target. The waiting time t_w starts at the first acquisition with the calibrated resistance-temperature detector (RTD) at the cell reading the target. Acquisitions followed one another every 6 s, and consecutive acquisitions were averaged in groups of about 100, each spanning about 10 min. Every t_w quoted below is the midpoint of a group (SI Appendix, section 6.1).

Fig. 2 follows the isothermal experiment at 28.50 °C through the hour in which the sample arrests. It shows seven groups recorded between $t_w = 300$ and 3740 s: the SAXS intensity

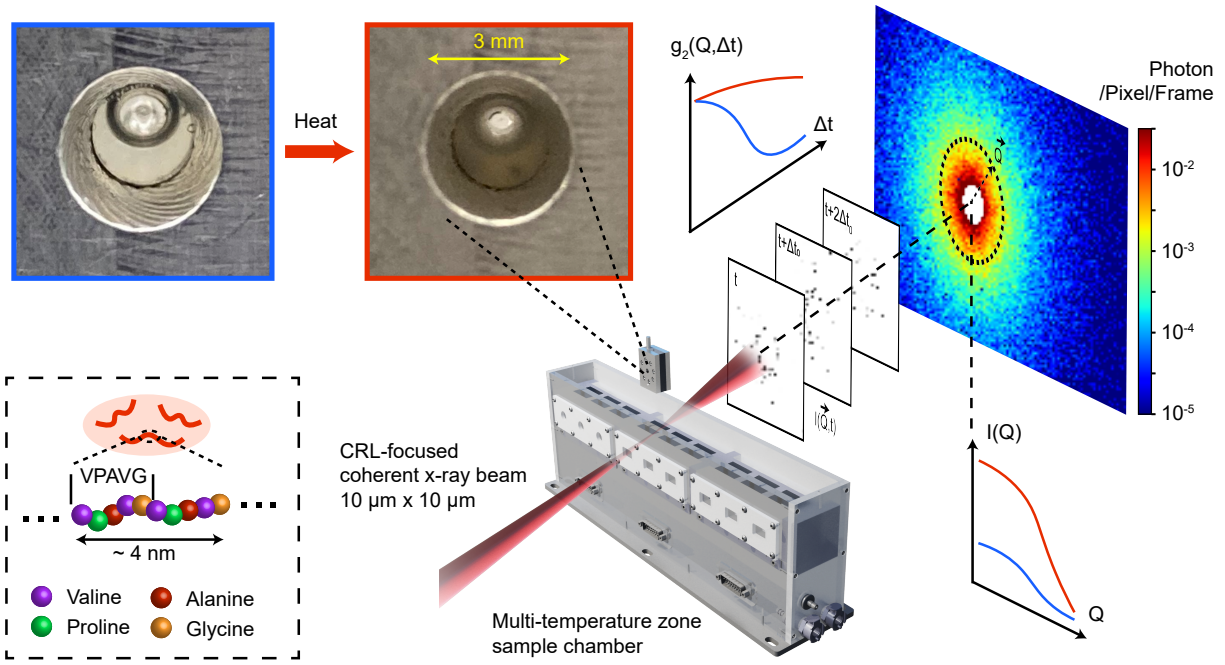


Figure 1: SA-XPCS measurement of thermally induced $(\mathrm{VPAVG})_{30}$ assembly at APS 8-ID-I. Each 100,000-frame set of sparse 20 µs frames feeds two analyses. Averaging them in time gives the SAXS intensity $I(Q)$: how much structure the sample has at each length scale $2\pi/Q$, which grows at low Q on heating. Correlating them in time gives $g_2(Q, \Delta t)$: how fast the structure at that same length scale rearranges. A decay in g_2 means the density fluctuations at $2\pi/Q$ are still moving; a flat g_2 means they are frozen over the accessible delay times Δt. Both are sketched beside the detector (red: assembled; blue: dispersed; illustrative, not data). Upper left: the aluminum cell, liquid chamber 3 mm across, photographed with the ELP sample below (blue outline) and above (red outline) the transition. Lower left: one $(\mathrm{VPAVG})_{30}$ repeat, residues colored as keyed. Right: the frame-averaged detector image, color bar in photons per pixel per frame, white center the beamstop mask; the graininess is sparse counting statistics, not resolved speckle.

(Fig. 2a), the correlation function at the largest length scale (Fig. 2b), and the parameters of the dynamics fitted at five length scales (Fig. 2c). The isothermal experiments at the other three temperatures are shown in the same format in SI Appendix, Figs. S4, S5 and S7. Radiation damage was controlled in two ways. Every acquisition used a previously unexposed sample position, so the cumulative dose does not grow with t_w, and the evolution in Fig. 2 cannot be a dose artifact. In addition, a control at 22 to 41 times the flux of this experiment showed no change in the SAXS or in g_2 with dose rate (SI Appendix, section 5 and Fig. S3). The three panels of Fig. 2 support three observations at this temperature, which we take in turn. The protein assembles on heating, its dynamics arrest once the growth of the assembled structure has slowed, and a two-mode analysis of the correlation functions measures how far that arrest has gone at each length scale.

First of all, the protein assembles on heating, and its structure keeps growing at low Q throughout the isothermal experiment (Fig. 2a). At 6 °C the background-subtracted intensity, on the absolute scale calibrated in SI Appendix, Fig. S2, is flat at about 0.03 mm^{-1}, as expected for dissolved protein. At 28.50 °C it is two to three orders of magnitude higher at low Q. The profiles stay broad and monotonic with no peak emerging, so the assembly gains density contrast without settling on a single characteristic domain size. We follow its progress through the low-Q intensity, the mean of $I(Q)$ over 0.004 to 0.008 $\mathrm{Å}^{-1}$ (shaded in Fig. 2a). This band spans the length scales of the correlation analysis below, and it is where the intensity grows most: over the isothermal experiment, $I(Q)$ rises 5.2-fold at 0.004 $\mathrm{Å}^{-1}$ but only 2.1-fold at 0.02 $\mathrm{Å}^{-1}$. In the first group at the target the low-Q intensity is already 76 times its level at 20 °C, and it grows fourfold more by the end of the isothermal experiment.

Secondly, arrest trails assembly: the dynamics arrest once the growth of the structure has slowed (Fig. 2b). At the largest length scale, $Q = 0.00359$ $\mathrm{Å}^{-1}$ (175 nm), the correlation functions decay in two steps: a relaxation on the millisecond scale, resolved only by the 50 kHz framing of the XSPA-500k,[23] followed by a plateau that persists through the 2 s acquisition. Such two-step relaxation, with a slow component that grows in time, is characteristic of

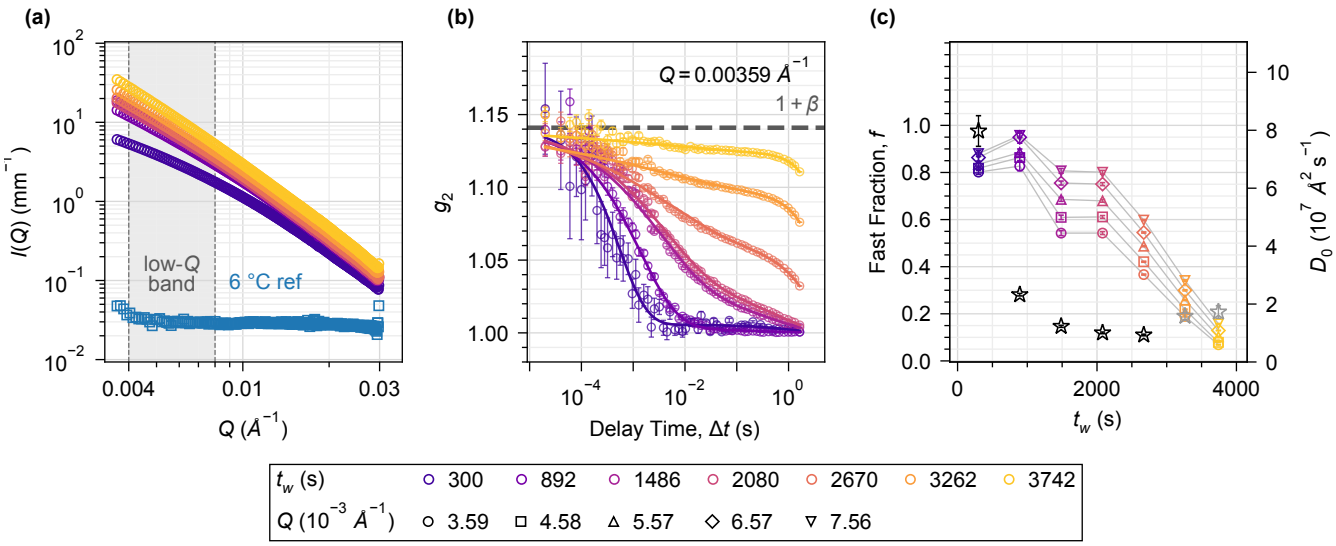


Figure 2: Assembly and dynamic arrest of $(\mathrm{VPAVG})_{30}$ at 28.50 °C. During the isothermal experiment the mesoscale structure keeps growing while the dynamics stop: the low-Q intensity rises fourfold while the fast fraction falls from 0.80 to 0.07 at the largest length scale. (a) Background-subtracted SAXS on the absolute scale of SI Appendix, Fig. S2, with the empty cell subtracted as in SI Appendix, section 6.2 and the 6 °C reference of the same loading as blue squares; profiles are drawn from the lowest fitted Q bin outward, and the shaded band, 0.004 to 0.008 Å^{-1}, is the low-Q band averaged in Fig. 3c. (b) g_2 at $Q = 0.00359$ Å^{-1} with fits of Eq. 2 at $\beta = 0.141$, p_{fast} free in each group; the dashed line is $1 + \beta$. (c) Left axis: the fast fraction f at the five Q values, marker shape denoting Q, grey lines guides to the eye. Right axis, stars: the fast-mode diffusion coefficient D_0, set by the 1/e time $1/(D_0 Q^2)$ of the fast decay and independent of p_{fast}; open grey stars mark groups with $f < 0.25$ at the lowest Q, where the fast decay is too weak to fix a rate. The rate falls threefold early in the experiment and then levels off while the amplitude keeps falling. The four remaining Q values are in SI Appendix, Fig. S6. Each t_w is the midpoint of a group averaging up to 100 acquisitions at fresh positions. Error bars are one-standard-deviation fit uncertainties; colour denotes waiting time, keyed along the bottom.

heterogeneous kinetic arrest in protein and colloidal gels.[5,24,25] In the first two groups, at 300 and 892 s, the first step carries about 80% of the decay, while the low-Q intensity doubles from 3.1 to 6.1 mm^{-1}. In the groups that follow, the first step shrinks and the plateau rises toward $1 + \beta$, the level of a sample that is fully arrested over the measured delays. Over these 2850 s the low-Q intensity grows only 2.1-fold, to 12.7 mm^{-1}, having already reached 147 times its level at 20 °C by 892 s.

Thirdly, a two-mode analysis of these correlation functions shows the arrested fraction growing at every length scale, first and most strongly at the largest, while the rate of the fast relaxation falls early and then levels off (Fig. 2c). The two-step decays were fitted with

$$
\begin{aligned}
g_2(Q,\Delta t,t_w) &= 1+\beta\,|g_1(Q,\Delta t,t_w)|^2\,,\\
g_1(Q,\Delta t,t_w) &= f(Q,t_w)\,g_{1,\mathrm{fast}} + [1-f(Q,t_w)]\,g_{1,\mathrm{slow}},\\
g_{1,\mathrm{fast}} &= \exp\!\left[-\left(D_0Q^2\Delta t\right)^{p_{\mathrm{fast}}}\right],\\
g_{1,\mathrm{slow}} &= \exp\!\left[-\left(\Delta t/\tau_{\mathrm{slow}}\right)^{p_{\mathrm{slow}}}\right].
\end{aligned}
\tag{2}
$$

The first line is the Siegert relation, in which g_1 is the normalized intermediate scattering function and β is the instrumental speckle contrast, measured on a static reference and held fixed (SI Appendix, section 6.2). The fast fraction f is the share of the density fluctuations at $2\pi/Q$ that relax on the fast time scale. Its complement, $1 - f$, is the share that stays arrested over the delays measured, so f falls from near 1 toward 0 as a sample arrests. The fast relaxation time is written as $1/(D_0Q^2)$, the form expected if the material that is still mobile moves diffusively. At that delay the fast term has decayed to 1/e of its amplitude, whatever the value of p_{fast}, so D_0 measures the rate of the fast relaxation. The stretching exponents p_{fast} and p_{slow} measure how broad each relaxation is, $p = 1$ being a single exponential. τ_{slow} lies beyond the acquisition window and is an extrapolated descriptor, so the analysis rests on f and D_0.

The fit uses the five lowest Q bins, 0.00359 to 0.00756 Å^{-1} or 175 to 83 nm; above them the scattered intensity has fallen by more than an order of magnitude and g_2 is dominated

by counting noise. In each group, f and τ_{slow} are free in every bin, while D_0, p_{fast} and p_{slow} are shared across the five. p_{fast} is free from one group to the next because it falls from about 1 in a mobile sample to about 0.2 in an arrested one, and a single value shared across an isothermal experiment is rejected by an F test (SI Appendix, section 6.3). The Q^{-2} form of the fast time is imposed rather than fitted. Released from it, the fitted exponent has a median of -2.5, and the collapse of Fig. 3 survives (SI Appendix, section 4.3).

With this model, Fig. 2c shows the fast fraction falling at every measured length scale; the correlation functions and fits at the four higher Q values are in SI Appendix, Fig. S6. At $t_w = 300\,\mathrm{s}$, f is 0.80 at 175 nm; by 3740 s it has fallen to 0.07, so the arrested fraction $1 - f$ has gone from 0.20 to 0.93 in under an hour. At every waiting time f also increases monotonically with Q, in all seven groups, so arrest develops first and most strongly at the largest length scale probed, 175 nm, and progressively less so toward 83 nm. That ordering holds within each group, so it reflects length scale rather than the trend with t_w. It ties the growing low-Q SAXS structure to the arrest measured on the very same frames.

The rate of the fast relaxation, shown as stars in Fig. 2c, behaves differently from its amplitude. While the sample is still largely mobile, D_0 falls from 8.0 to 2.3×10^7 $\mathrm{Å}^2\,\mathrm{s}^{-1}$. Once f has dropped below 0.7, D_0 stays between 0.9 and 1.2×10^7 $\mathrm{Å}^2\,\mathrm{s}^{-1}$ while $1 - f$ climbs from 0.46 to 0.63, and the amplitude keeps falling, to 0.07, after the fast decay has become too weak to fix a rate (open stars). The longer isothermal experiment at 28.25 °C shows the same over a wider range: the rate stays within a factor of 1.7, with no trend, while f falls from 0.67 to 0.29 (SI Appendix, Fig. S8). The same pairing of a plateauing fast time scale with a declining fast amplitude was found in a thermoreversible nanoparticle gel.[24] The fast relaxation also broadens, p_{fast} falling from 0.7–0.9 at the start to 0.20–0.35 at the end of each isothermal experiment in which the sample arrests. At $p_{\mathrm{fast}} = 1$ the fast term falls from 90 to 10% of its amplitude over 1.3 decades of delay; at 0.2 it takes 6.7 decades. Late in arrest, therefore, the amplitude of the fast relaxation keeps falling, while its rate stays nearly constant and its decay spreads over a widening range of time scales.

Fig. 3 extends the analysis to all four isothermal experiments, at 28.00, 28.25, 28.50 and 28.75 °C, and shows that one temperature-scaled clock paces both the arrest and the assembly. It follows f at the largest length scale, $Q = 0.00359$ Å^{-1} (175 nm), where arrest goes furthest, in all 34 fitted groups. Fig. 3a plots this f against t_w, Fig. 3b plots it against a rescaled waiting time, and Fig. 3c places the low-Q intensity of the same isothermal experiments on that rescaled time.

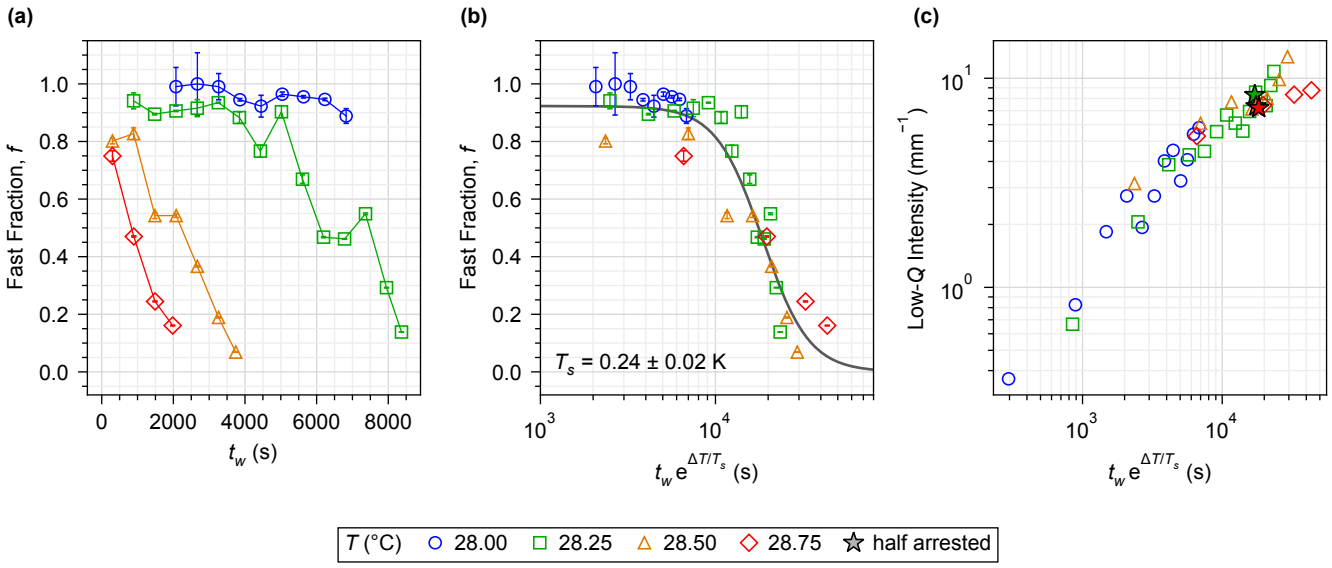


Figure 3: How fast arrest runs, and the single rescaling that collapses it at every temperature. (a) The fast fraction f at the largest length scale, $Q = 0.00359$ Å^{-1}, against t_w, the midpoint of each group. At 28.00 °C f stays at 0.89 or above for 6800 s; at the three higher temperatures it falls, and the fall begins earlier as the temperature rises. (b) The same f against $t_w\, e^{\Delta T/T_s}$, with $\Delta T = T - T_0$. The grey line is one master curve, $f = A/[1 + (X/X_c)^n]$, fitted to all four isothermal experiments with T_s free; it improves the residual threefold over no rescaling. $T_0 = 28.00$ °C is a convention that slides the curve without changing T_s, whose uncertainty is a bootstrap standard deviation over groups. The collapse is approximate: the curve lies about eight fit uncertainties from the points. (c) The low-Q intensity, $I(Q)$ averaged over the band shaded in Fig. 2a (0.004 to 0.008 Å^{-1}), of every group of the four isothermal experiments against $t_w\, e^{\Delta T/T_s}$ with the T_s of (b). Stars mark where each sample that arrests is half arrested, where f in (a) falls to 0.5. Colour and marker shape denote temperature, keyed along the bottom; error bars are one-standard-deviation fit uncertainties.

To begin with, the rate of arrest depends steeply on temperature (Fig. 3a). At 28.00 °C, f stays at 0.89 or above throughout the 6800 s isothermal experiment, although the protein assembles (SI Appendix, Fig. S4). At the three higher temperatures f falls, and the fall sets

in earlier at higher temperature: f reaches 0.5 after 6100, 2220 and 830 s at 28.25, 28.50 and 28.75 °C, a 7.3-fold change over half a degree.

Next, one rescaling of the waiting time collapses the arrest at all four temperatures (Fig. 3b). Against $X = t_w \exp[(T - T_0)/T_s]$, f falls onto one master curve, $f = A/[1 + (X/X_c)^n]$, with $T_s = 0.24 \pm 0.02$ K. Collapsed one Q bin at a time, T_s reads 0.24 to 0.27 K, so the same temperature scale holds at every length scale measured, and the time scale of the master curve grows 1.3-fold from 175 to 83 nm (SI Appendix, Fig. S9). The collapse says more than that arrest is faster at higher temperature: one factor on the waiting time accounts for most of the difference between the four experiments, so temperature sets the pace of arrest. The 28.00 °C sample, which never arrests during its isothermal experiment, is the same curve seen too early: on the master curve it would be half arrested near 17 600 s. A scaling temperature of 0.24 K means that arrest speeds up about threefold for every quarter of a degree of heating. A dependence this steep indicates a cooperative step, in which many contacts form together, as discussed below.

Finally, the assembly runs on the same clock (Fig. 3c). Fig. 3c places the low-Q intensity of every group, fitted or not, on $t_w \exp[(T - T_0)/T_s]$ with the T_s of Fig. 3b. The intensities of all four isothermal experiments fall on one curve, and fitted on their own they return the same $T_s = 0.24$ K. Other bands tested between 0.003 and 0.03 Å^{-1} superpose at 0.24 or 0.25 K (SI Appendix, section 6.5). Structure and arrest therefore advance on a common clock.

Along that clock, assembly runs ahead of arrest. To compare the two, we mark each sample as half arrested when f at 175 nm falls to 0.5, interpolated between the two groups that bracket it (stars in Fig. 3c). At that point half of the density fluctuations at 175 nm still relax on the fast time scale and half are arrested. It lies near the midpoint of the fall of f, and all three samples that arrest pass it. Each is half arrested when its low-Q intensity reaches 7.2 to 8.3 mm^{-1}, 175 to 220 times its level at 20 °C, and the intensity grows only 1.2- to 1.7-fold more while f falls from 0.5 to between 0.07 and 0.16. The isothermal experiment

at 28.00 °C ends at 5.8 mm^{-1} with f still 0.89, short of that stage. Other thresholds give the same picture: for any value between $f = 0.3$ and 0.7, the samples cross it at 5.7 to 9.2 mm^{-1}, and the intensity grows 1.1- to 1.9-fold afterwards (SI Appendix, section 6.5). Arrest therefore follows assembly: a sample is half arrested only after most of its low-Q structure has formed.

Discussion

The Introduction asked whether the arrest of a protein assembly simply tracks the growth of its structure, and what sets its pace. Measuring structure and dynamics on the same frames answers both questions for $(\mathrm{VPAVG})_{30}$. Arrest follows assembly: the low-Q structure forms first, and the dynamics arrest as its growth slows. Both nonetheless advance on one clock, and temperature sets its pace. That clock is unusually steep, speeding up about threefold for every quarter of a degree of heating. Below we discuss what so steep a clock implies about the step that sets it, how that step fits with what is known about ELPs, and what it means for protein materials and condensates.

First, the steepness of the clock points to a cooperative step. Written as an Arrhenius rate, $k \propto \exp(-E_a/k_\mathrm{B}T)$, a scaling temperature of 0.24 K corresponds to an apparent activation energy $E_a = k_\mathrm{B}T^2/T_s \approx 1.2 \times 10^3\, k_\mathrm{B}T$, or about $3\,\mathrm{MJ\,mol}^{-1}$. An energy that large must be shared among many contacts. The same collapse was introduced for thermoreversible colloidal gels, where the formation rate varies with quench depth far more steeply than the pair attraction alone allows, and that steepness was attributed to the cooperative bonding needed to build locally favored structures.[26,27] In those gels, $T_s = 0.33$ and 0.45 K for two particle sizes. Because the temperature dependence of the pair attraction was known, these values gave local structures held by about 26 and 19 bonds.[24,26] In a dense protein phase, whose contacts lie far below optical resolution, the scaling temperature is thus an in situ kinetic measure of how many contacts must form together.

Protein assembly gated by nucleation shows the same kind of steepness. The delay before

deoxyhemoglobin S gels on heating varies as about the 30th power of protein concentration and almost halves for every degree of warming, which identified the nucleation of fibers as the rate-limiting step.[28] How steeply such a rate depends on concentration counts the molecules in the critical nucleus,[29] and protein aggregation more generally accelerates once nucleation has started.[30] For an ELP, heating acts much like concentrating. ELPs assemble sharply once heated above their transition temperature,[9,10] and at a fixed temperature a change in concentration can trigger the same transition.[8,13] The arrest clock here speeds up about sixtyfold per degree, against twofold for sickle hemoglobin. For this protein, the temperature dependence of a single contact is not calibrated, so no bond number is extracted. The scaling still requires that arrest be gated by a step that forms many contacts at once.

Second, the step that sets the clock builds the mesoscale structure. Above the LCST, ELP assembly is driven by hydrophobic desolvation, and desolvation is well under way before any isothermal experiment begins: the low-Q intensity has already grown two- to fourfold by 27.5 °C on the ramp (SI Appendix, section 6). The equilibrium of a single contact shifts very little over half a degree, yet half a degree changes the rate of arrest sevenfold, and the low-Q structure grows on the same clock. The rate-limiting event is therefore one that builds the mesoscale structure and then freezes it: the concerted formation of junctions in which many contacts close together.

Both mechanisms raised in the Introduction can supply such a step. In associative networks with ELP end-blocks, the end-blocks aggregate into junctions whose association and dissociation set the network relaxation time,[20] so forming or breaking one junction moves a whole ELP block, many residues together. In a gel of associating coiled-coil proteins, tracer diffusion likewise suggests that clusters of chains move together on large length scales and detach from the network only when several coiled-coil domains dissociate.[31] In viscoelastic phase separation, a deep quench turns the slow, protein-rich phase into a transient gel and stalls its coarsening, and near a glass transition the mobility of that phase drops steeply.[21] Relaxation there requires many molecules to rearrange together.[32] For an LCST protein, a

deeper quench is a higher temperature. In both pictures, the step that sets the clock involves many contacts or molecules acting at once, which is what the steep temperature dependence requires.

Third, what is known about ELPs supports this picture. Individual contacts in an ELP coacervate are short-lived. NMR shows transient and nonspecific hydrophobic contacts between disordered chains,[33] and simulations find the chains hydrated and highly disordered even as assembly triples their nonpolar side-chain contacts.[34] A structure that persists for hours must therefore be held by many such contacts at once. ELPs that carry alanine in the third position of the repeat, as $(\mathrm{VPAVG})_{30}$ does, are also known to arrest. Concentrated solutions of $(\mathrm{XPAVG})_n$ ELPs warmed above their transition temperature undergo arrested phase separation into stiff gels, and related ELPs form bicontinuous networks with stress-relaxation times beyond 10^3 s.[16,17] Alanine-containing ELP blocks also exchange between junctions more slowly than glycine-containing ones,[20] and poly(VPAVG) loses much of its structural disorder as its coacervate matures.[9] The sequence observed here is consistent with network-forming arrested phase separation in a dynamically asymmetric protein–solvent mixture, where concentration fluctuations build a connected dense phase and viscoelastic constraints suppress relaxation.[5,18,21,35] These studies characterize the arrested state. The present data add how fast it is reached: arrest follows assembly, on a clock set by a cooperative step.

Finally, the steepness of the clock matters in practice. At $T_s = 0.24$ K, a quarter of a degree of warming shortens the time to arrest by a factor of 2.8, so the rate at which an ELP network assembles and arrests is adjustable over orders of magnitude by sub-degree temperature control.

What paces arrest is also an open question for biomolecular condensates, whose maturation from liquid to solid governs their function and their role in aggregation.[36,37] Condensates are too small and too dilute for the measurement reported here, but a concentrated, sequence-defined mimic is not. In this mimic, arrest follows assembly on a common clock, and the steep

temperature dependence of that clock points to a cooperative step. Whether condensates harden through a step of the same kind can be tested with the tools already used on them: if they do, their arrest followed at a few temperatures should collapse under one rescaling of time.

Two limitations bound the interpretation, and each points to the measurement that would lift it. First, the 2×10^{-5}–1.6 s delay range quantifies the fraction arrested on that scale rather than the infinite-time nonergodicity factor, for which linear shear rheology on a scaled-up preparation is the complement. Second, the scaling rests on three arresting temperatures spanning half a degree, and the collapse is approximate: rescaling improves the residual threefold but not to the measurement uncertainty. More temperatures would settle it. T_0 is a convention rather than a measured transition value, so no reference time is quoted from it.

Subject to those constraints, the measurements give one picture. SA-XPCS follows what SAXS alone cannot: how far an assembly has arrested. Structure and arrest run on one clock, with arrest following assembly and finishing after most of the low-Q structure has formed. Arrest reaches the largest length scales first. One rescaling of time brings every isothermal experiment onto one master curve, so temperature sets the pace of arrest. That clock runs almost three times faster for every quarter of a degree of heating, a steepness that points to a rate-limiting step in which many contacts close together.

Materials and Methods

The Sanger-verified construct encoded an N-terminal His-tagged $(VPAVG)_{30}$ protein (complete sequence in SI Appendix). Recombinant $(VPAVG)_{30}$ was expressed in *E. coli*, purified by immobilized metal-affinity chromatography, and buffer exchanged into 20 mmol L^{-1} HEPES, 250 mmol L^{-1} NaCl, and 2 mmol L^{-1} DTT at pH 8. Expression, solubility, and purity were monitored by SDS-PAGE (SI Appendix, Fig. S1b). Protein concentration was determined from A_{280} on a NanoDrop spectrophotometer with $E_{280}^{1\%} = 0.95$; the batch used for all four

loadings was at $101\,\mathrm{mg\,mL^{-1}}$.

SA-XPCS was performed at APS 8-ID-I with a 10.94 keV, 10×10 $\mu\mathrm{m}^2$ partially coherent beam and a Rigaku XSPA-500k detector 7.8 m downstream. Each acquisition contained 100,000 frames at 50 kHz (20 µs per frame, 2 s total), and the four isothermal experiments were carried out in one beamtime in a single cell block. Further acquisition, fitting and radiation-control details are in SI Appendix.

The four loadings were measured one after another on the same temperature zone and resistance-temperature detector, in the order 28.00, 28.50, 28.25 and 28.75 °C, and the detector reading varied by 0.004 °C (standard deviation) during each isothermal experiment. Acquisitions followed one another every 6 s, each at a fresh position. Consecutive acquisitions were averaged in groups of about 100 after two outlier cuts (SI Appendix, section 5.3), and each group is placed at the midpoint of the acquisitions it contains, $t_w = 0$ being the first acquisition with the detector at its target. Eq. 2 was fitted to the five lowest Q bins of each group at once by error-weighted least squares, with β fixed at the value measured on a static reference, f and τ_{slow} free in each bin, and D_0, p_{fast} and p_{slow} shared across the bins of a group (SI Appendix, section 4).

For the scaling analysis, f at the lowest Q of every group of the four isothermal experiments was fitted with one master curve, $f = A/[1+(X/X_c)^n]$ with $X = t_w \exp[(T-T_0)/T_s]$, weighting each point by its fit uncertainty floored at 0.01, and each higher Q bin was collapsed the same way on its own. $T_0 = 28.00\,^{\circ}\mathrm{C}$ is a convention: it moves the curve along X without changing T_s. The uncertainty of T_s is the standard deviation over bootstrap resamples of whole groups. The low-Q intensity is the mean absolute $I(Q)$ of each group over 0.004 to 0.008 $\mathrm{Å}^{-1}$. A sample is half arrested at the waiting time at which f at the lowest Q falls to 0.5, interpolated linearly between the two groups that bracket it (SI Appendix, section 6.5).

Data, Materials, and Software Availability

Reduced data and analysis/figure-generation code are publicly available at `https://github.com/qzhang234/PubCode-2021-IDPP` [**archived at Zenodo; DOI to be added**]. The repository includes the 8-ID-I datasets used for the figures, radiation-control data, and analysis scripts, and every data-derived figure can be regenerated from it alone. Raw beamline data are available from the corresponding authors upon reasonable request.

Author Contributions

[**To be completed by the authors.**]

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AI-assisted manuscript and code preparation. The authors used Anthropic's Claude models, accessed through Argo, Argonne National Laboratory's internal generative AI interface, to assist with manuscript preparation and with development of the data-analysis and figure-generation code (see SI Appendix). That code and the reduced data it operates on are publicly available at `https://github.com/qzhang234/PubCode-2021-IDPP`, so the reported results can be regenerated and checked independently. The authors take full responsibility for all manuscript content and reported results.

Competing Interest Statement

The authors declare no competing interest.

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